

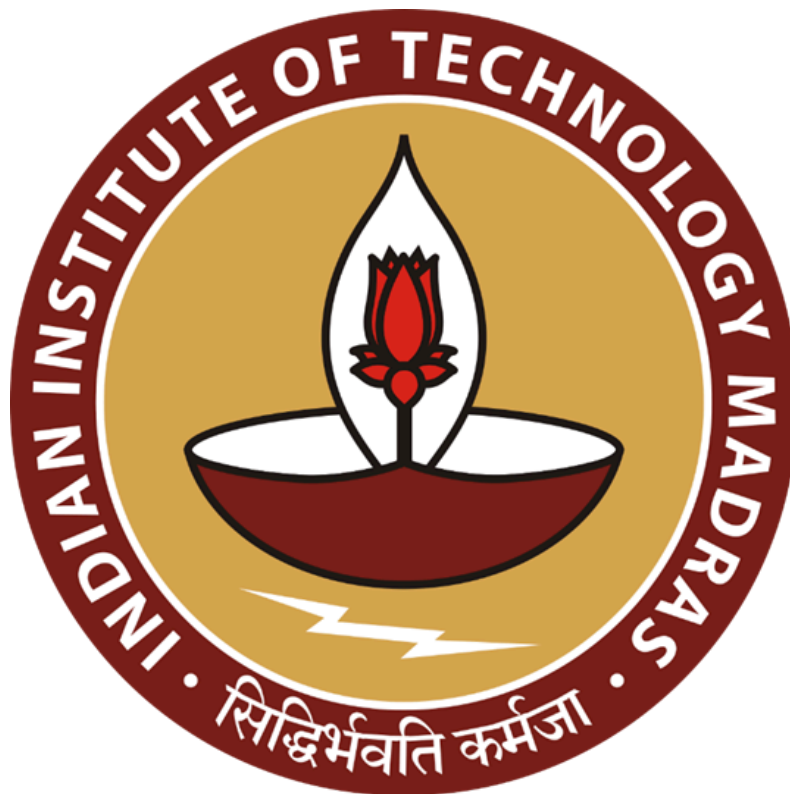
**Optimizing B2B Operations through Credit Risk and Inventory Analytics at Ace and
Bees Healthcare Devices**

Final Report for the BDM Capstone Project

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Executive Summary

Ace and Bees Healthcare Devices is a small B2B medical equipment trading firm operating from Cochin and Bangalore, facing recurring challenges related to delayed customer payments and rapid growth in inventory levels. These issues have placed sustained pressure on working capital and reduced clarity in financial decision-making. This project was undertaken to analyse the firm's credit exposure and inventory structure using internal operational data and to provide insights for improving financial discipline and liquidity management.

Four datasets were obtained from the firm: the sales register, sales aging report, balance sheet debtor list, and inventory records. Data were cleaned, standardised, and anonymised to ensure accuracy and confidentiality. Credit aging analysis, client risk scoring, inventory classification, and descriptive statistics were applied using spreadsheet-based tools to convert raw business data into actionable insights.

The analysis shows that overdue payments are highly concentrated among a small number of clients, with several accounts carrying significantly delayed balances. Inventory results indicate that a small set of products accounts for most of the stock value, along with a sharp increase in total inventory over a short period. These patterns point to a mismatch between cash inflows and capital deployment, where funds remain blocked in both unpaid receivables and unsold stock.

Based on the findings, the project recommends stronger credit control, prioritised collection efforts, and closer alignment between procurement and sales activity. If implemented, these measures may improve liquidity and support more stable business operations.

2. Detailed Explanation of Analysis Process / Method

The project focuses on two core operational issues at Ace and Bees Healthcare Devices: delayed client payments and rapid, concentrated growth in inventory. The analysis process was designed to use the firm's own records to quantify these problems and provide decision-ready insights that the business can realistically implement. This section explains the data used, the cleaning and preprocessing steps, the methods applied for credit and inventory analysis, and the rationale for choosing these methods.

2.1 Data sources and link to problem statements

Four datasets were obtained directly from the firm's accounting and stock records:

- The **Sales Register** contained invoice-level data, with one row per sales transaction, including invoice date, party name, transaction amount, received amount and balance due. This dataset is central to **Problem Statement 1: credit risk and delayed payments**.
- The **Sales Aging Report** recorded due dates, days late and overdue amounts for each invoice and supported the analysis of payment delays and aging patterns.
- The **Debtor List from the Balance Sheet** provided party-wise outstanding balances and was used to cross-check receivables and assess their impact on the firm's short-term financial position.
- The **Inventory Records** contained SKU-wise data on quantity, unit rate and total value at different points in time and formed the basis of **Problem Statement 3: inventory concentration and stock build-up**.

The sales register contained 60 invoices, representing all sales transactions that the firm shared for this project. Since Ace and Bees is a small B2B firm dealing in relatively few but high-value orders, this dataset reflects its actual transaction pattern. The methods were therefore designed to extract maximum insight from a compact, real-world dataset.

2.2 Data cleaning and preprocessing

Before any analysis, the raw data from these different files had to be made consistent, accurate and usable together.

The first step was **format standardisation**. Dates related to invoice issue and due dates were converted into proper date formats so that time differences could be computed correctly. All monetary fields—such as total amount, balance due and stock value—were expressed in Indian Rupees (₹) with two decimal places. Text fields like party names and product names were cleaned for extra spaces and inconsistent capitalisation.

The second step involved **handling missing and inconsistent values**. Records where critical fields, such as total amount or balance due, were missing or clearly incorrect were reviewed. Where they posed a risk of distorting totals or averages, they were excluded from the relevant quantitative summaries. Non-critical gaps, such as missing remarks, were left as they were, to avoid unnecessary alteration of the original records.

Client names were first standardised to correct spelling differences so that invoices belonging to the same client could be aggregated accurately. For confidentiality, **only clients with outstanding or defaulted payments were anonymised** in the report and replaced with neutral labels such as **Party A, Party B**, and so on.

Finally, several **derived variables** were created. For each invoice, a **Status** field (Paid or Unpaid) was computed from the balance due. A **Days Late** field was calculated using a fixed reference date; in simple form:

$$\text{Days Late} = \text{Reference Date} - \text{Due Date}$$

For inventory, the **Value** of each item was calculated as:

$$\text{Value} = \text{Quantity} \times \text{Rate}$$

and each item's **Value Share** in total inventory was:

$$\text{Value\%} = \text{Value} / \text{Sum of all Values}$$

These derived columns act as building blocks for later steps like aging analysis, ABC classification and risk scoring.

Importance of data cleaning.

Without these cleaning and preprocessing steps, the analysis would be vulnerable to basic errors—wrong dates leading to incorrect aging, inconsistent party names splitting a single client into several entries, or missing amounts distorting average delays and balances. Cleaning therefore ensures that the patterns identified in later sections reflect real business behaviour rather than data noise. It also makes the analysis replicable for the firm in future months.

2.3 Descriptive statistics and summarisation

As part of the analysis, basic descriptive statistics were computed for key financial variables in order to understand the underlying structure of the dataset before applying any classification or risk-scoring models. Specifically, the mean, median, minimum, maximum and standard deviation were calculated for invoice amounts, balance amounts, and days late. These measures were used to examine the typical size of transactions, the extent of variation in payment delays, and the level of concentration in outstanding dues. Similar descriptive measures were also computed for inventory values to assess the degree of value concentration across stock items. Descriptive statistics also helped identify whether observed patterns were driven by a few extreme cases or reflected general behaviour across invoice records. This ensured that subsequent analysis, such as risk categorisation and inventory classification, was grounded in an accurate summary of the firm's transactional behaviour.

2.4 Credit-risk analysis methods

The credit-risk analysis aims to translate the raw invoice and aging data into a clear picture of which clients pose the greatest threat to cash flow.

Party-wise exposure.

Using pivot tables on the sales register, invoices were aggregated by party to obtain, for each client, the total billed amount and the total outstanding balance. This allows identification of clients with both high sales volume and high unpaid balances, directly addressing concerns about delayed payments and credit concentration.

Aging buckets.

To see how long payments had been overdue, each invoice's Days Late was grouped into aging buckets such as 0–30, 31–60, 61–180 and over 180 days. The share of total overdue

amount in each bucket was then calculated. This helps distinguish between short, routine delays and serious, long-term arrears.

Formally, for each invoice i , Days Late determines the bucket into which it is assigned. The total overdue value in a bucket is the sum of balances for all invoices in that range, and dividing by the overall overdue value gives the percentage share.

Forecasted collections.

Because not all overdue balances are equally likely to be recovered, a simple rule-based estimate of **forecasted collections** was created. For each client, the model assigns a proportion of their outstanding balance that is likely to be collected, based on how delayed their average payment is. For example, balances with relatively low average delay are treated as mostly collectible, while those with very high delay are treated as unlikely to be recovered. The remainder is treated as “unlikely to collect” for planning purposes.

While this is not a probabilistic model, it offers a transparent, adjustable framework the firm can work with, and it links directly to problem statement 1 by quantifying how much of the receivables are realistically usable.

Risk scoring and categorisation.

To combine delay and exposure into a single measure, a **risk score** was defined for each party. In conceptual form:

$$\text{Risk Score} = f(\text{Average Days Late}, \text{Balance})$$

where the function increases when either average delay or outstanding balance increases. Based on the risk score, each client was classified into low-, medium- or high-risk categories. These categories were then associated with recommended credit policies: low-risk clients may receive normal terms; medium-risk clients may receive shorter credit days; high-risk clients would be candidates for advance or cash terms only.

Link to liquidity.

Finally, the impact of receivables on overall liquidity was examined through the **current ratio**, defined as:

$$\text{Current Ratio} = \text{Current Assets} / \text{Current Liabilities}$$

The current ratio was computed for the present situation and for a scenario where forecasted collections are realised. This links credit risk directly back to the project's focus on working capital and financial stress.

2.5 Inventory analysis methods

The inventory analysis methods were chosen to show how much capital is tied up in stock, where that capital is concentrated, and which items create the most risk.

ABC analysis.

Using the value of each SKU, items were ranked from highest to lowest value, and each item's share of total inventory value was calculated. The cumulative share was then used to classify items into A, B and C categories. In broad terms, A-class items account for around 70–80 percent of stock value, B-class for the next band, and C-class for the remaining small share. This shows a small group of critical items that dominate inventory value.

Cost tiers.

Products were also grouped into cost tiers (low, medium, high) based on their unit rates. This helps management see whether most capital is tied up in a few very expensive items or whether it is more evenly spread out.

Stock risk flags.

To identify particularly sensitive items, the analysis flagged SKUs that were both high in value and low in quantity. These items are risky because they combine the possibility of stock-outs (too few units) with high capital blockage (each unit is expensive). Such flags support more careful monitoring and more cautious procurement.

Inventory growth over time.

Finally, total inventory value at two points in time was compared to see how fast stock was growing. The percentage change in total inventory value is given by:

$$\% \Delta \text{Inventory} = [\text{Inventory}(\text{later}) - \text{Inventory}(\text{earlier})] / \text{Inventory}(\text{earlier}) \times 100$$

In this project, this revealed a very sharp increase over a short period, reinforcing concerns about overstocking and weak demand-based planning.

2.6 Justification of methods and tools

The choice of spreadsheet-based methods, descriptive statistics, aging analysis, simple risk scores and ABC classification was intentional. Ace and Bees is a small firm without access to specialised analytics software or dedicated data teams. Google Sheets and Excel were therefore used as the primary tools, as they are already part of the firm’s routine operations and are easy to maintain.

The methods are directly tied to the problem statements. Aging analysis, party-wise exposure and risk scoring address delayed payments and credit concentration. ABC analysis, cost tiers and inventory growth address value concentration and overstocking. Mathematical abstraction is kept simple and transparent so that the firm can understand, question and adjust the parameters. Instead of complex models that require large datasets and statistical expertise, the project emphasises clarity, usability and direct business relevance.

In summary, the analysis process converts scattered operational records into a structured view of credit risk and inventory stress, using methods that fit the scale of the business and can continue to support decision-making after the project is complete.

3. RESULTS AND FINDINGS

This section presents the outputs of the credit-risk and inventory analyses and interprets them in relation to Ace and Bees Healthcare Devices’ operational challenges. Each subsection discusses visual outputs, statistical patterns, and business implications derived from the data.

3.1 Descriptive Statistics: Invoice and Receivable Patterns

Table 1: Summary of Descriptive Statistics – Invoice Data

Variable	Count	Mean	Median	Std.Dev	Minimum	Maximum
Invoice Amt(All)	60	3.07 L	81,228	6.65 L	1220	38.60 L
Days Late	10	420.7	165.5	621.8	2	1624
Balance Due	10	1.13 L	36,394	1.99 L	2927	6.38 L

For the complete dataset of 60 invoices, the average invoice value is ₹3.07 lakh, while the median is ₹81,228, indicating wide variation across transactions. The large standard deviation and the gap between the smallest and largest invoices confirm that although many invoices are small in size, a small number are extremely large in value.

Delay statistics were available for 10 invoices. The average delay of 420.7 days, with a maximum of 1,624 days, shows that payment delays extend far beyond routine late payments and include cases of long-term non-payment. Outstanding balances show similar skewness, with a mean of ₹1.13 lakh and a median of ₹36,394, indicating that receivable risk is driven by a small number of large overdues rather than many small ones.

These patterns confirm that the firm's credit exposure is not evenly spread but concentrated in a few problematic accounts, increasing its vulnerability to default and cash-flow disruption.

3.2 Descriptive Statistics: Inventory Value Distribution

Table 2: Summary of Descriptive Statistics – Inventory Value per Item

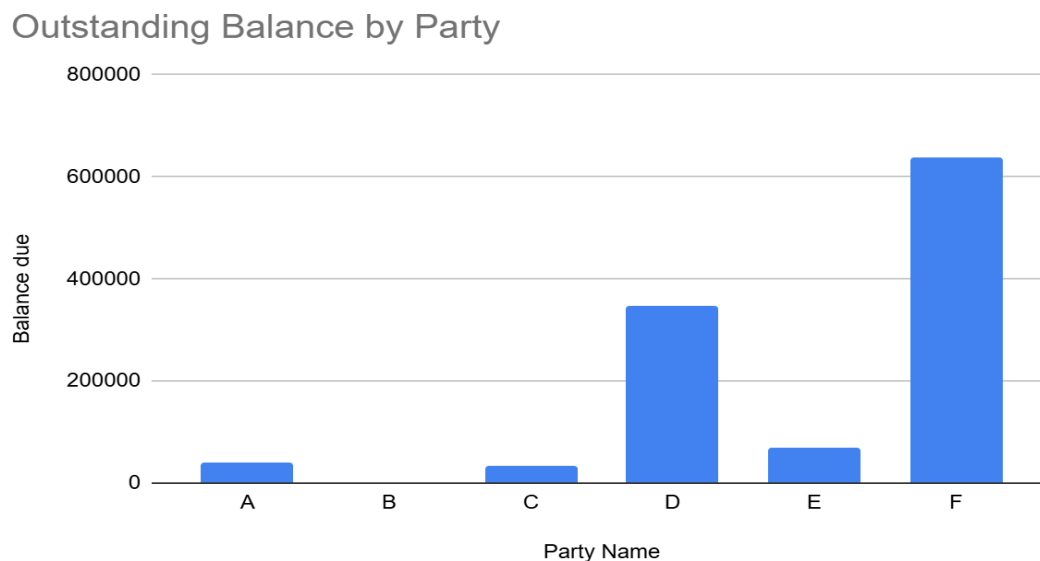
Variable	Count	Mean	Median	Std.Dev	Minimum	Maximum
Stock Values	26	2.05 L	7750	7.53 L	154	38.39 L

Descriptive statistics for inventory were calculated for 26 stock items. The average value per item is ₹2.05 lakh, while the median is only ₹7,750, showing that most inventory items are low in value but a few products account for extremely high values. This is confirmed by the very high standard deviation of ₹7.53 lakh.

The large difference between the minimum and maximum values suggests that capital is highly unevenly distributed across products. As a result, inventory risk is concentrated in a small number of high-value items. This explains the results of the ABC analysis and highlights the importance of focusing control efforts on a limited number of critical products rather than treating all stock items equally.

3.3 Concentration of Outstanding Receivables by Client

Figure 1: Outstanding Balance by Party

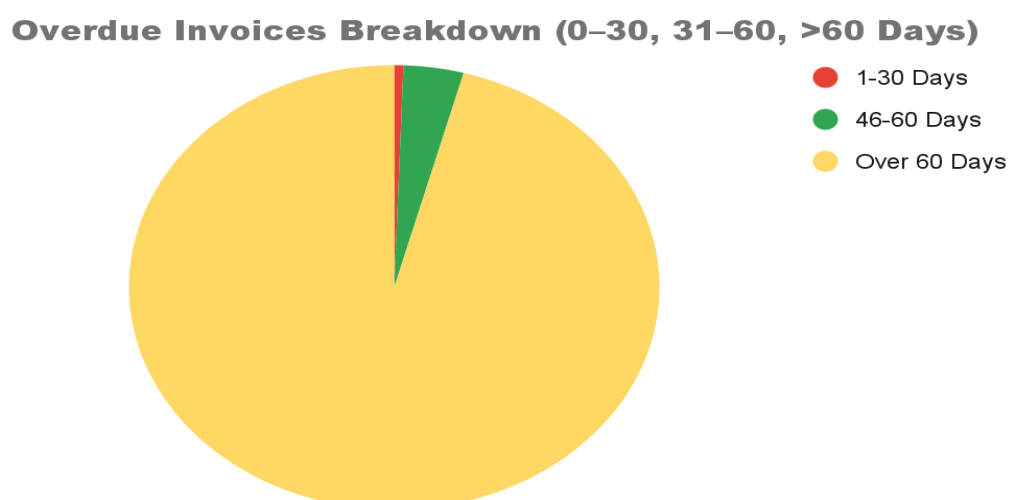


The distribution of outstanding receivables across clients is highly uneven. Two to three hospital clients account for a disproportionately large share of total dues, while the remaining clients contribute marginally. This imbalance indicates a **risk concentration problem**, where cash inflow is critically dependent on a small number of accounts.

Such concentration increases the firm's vulnerability: if just one large client defaults or delays heavily, liquidity can deteriorate sharply. Instead of risk being diversified across many small receivables, it is structurally concentrated, exposing the firm to systemic disruption from the behaviour of just a few customers.

3.4 Invoice Aging Pattern and Overdue Risk Distribution

Figure 2: Overdue Invoices Breakdown by Aging Bucket



The aging profile reveals severe payment delays. More than half of the unpaid invoices fall into the **over-60 days** category, with several invoices exceeding 180 days overdue. This pattern confirms that delayed payments are not temporary or isolated events but are structurally embedded in the firm's receivable cycle.

As invoice age increases, the probability of recovery falls. Therefore, the high share of invoices in advanced aging buckets implies escalating credit risk and weak enforcement of credit terms. The firm appears to be operating with credit policies that are insufficiently monitored and weakly enforced, allowing overdue balances to accumulate.

3.5 Results of Credit Risk Classification

Table 3: Client-wise Risk Category (Low / Medium / High)  Credit_Risk_Model

Party	Avg Days Late	Balance Amount (Aging)	Historic al Total Amount (Sales)	Balance from Balance Sheet	Risk Score	Risk Category	Recommendation
A	51.5	41020	80500	41020	26.00	Low	Extend credit
B	1624	2927	1708296	2927	812.00	High	Avoid credit; demand upfront payment
C	112	32446	32446	32446	56.50	Medium	Short credit terms only
D	181.666 6667	347626	544131	347626	91.15	High	Avoid credit; demand upfront payment
E	131	70560	108640	70560	65.82	High	Avoid credit; demand upfront payment
F	1561	637560	1912680	637560	780.67	High	Avoid credit; demand upfront payment

The credit-risk scoring model classifies clients into three categories: low, medium, and high risk. Four parties (B, D, E, and F) fall into the **high-risk category**, with exceptionally high average payment delays—ranging from over 130 days to more than 1,600 days in one case.

These accounts exhibit two dangerous characteristics simultaneously:

A. Large outstanding balances

B. Persistent delays in historical payments

This combination indicates not merely liquidity problems on the client side but also **policy failure** on the firm's side—credit continues to be extended to entities that are statistically unlikely to pay on time.

3.6 Impact of Receivables on Liquidity Position

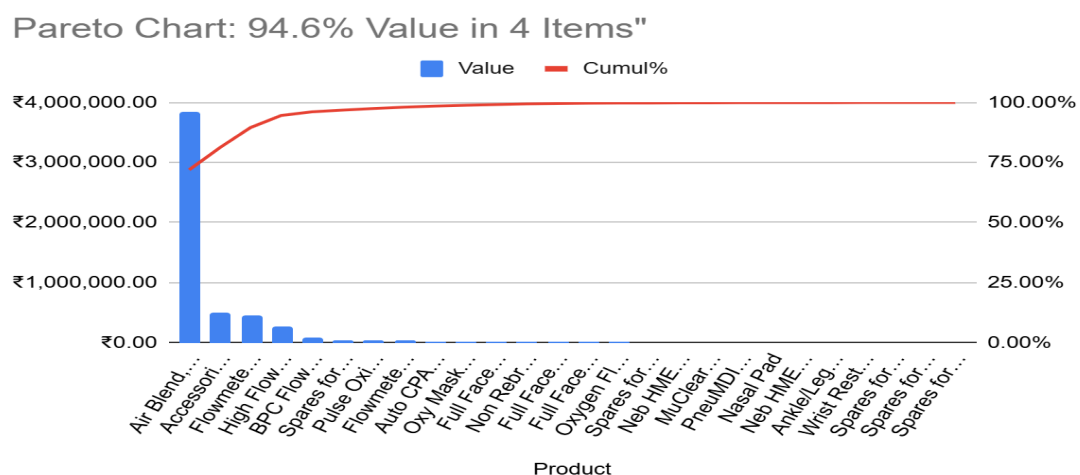
The firm's baseline current ratio is **0.81**, meaning short-term liabilities exceed short-term assets. This suggests liquidity stress even before accounting for potential defaults.

When forecasted collections are incorporated, liquidity improves marginally but does not fully stabilize. This confirms that the firm's liquidity weakness is not only due to non-payment but also due to overcapitalisation in inventory. In other words, even perfect collections would not fully solve the liquidity problem unless inventory practices are also corrected.

This dual constraint—**slow collections and rapid stocking**—creates a liquidity trap.

3.7 Inventory Concentration and ABC Classification

Figure 3: Pareto Chart – Inventory Value by SKU (ABC Classification)



The ABC analysis reveals an extreme concentration of inventory value, with only three to four SKUs accounting for nearly 90% of total stock value, while more than twenty items collectively contribute less than 5%. This indicates that the firm's profits, storage burden, operational risk, and capital lock-up are all heavily dependent on a very small subset of products. As a result, the company becomes financially sensitive to the performance of these critical items. If an A-class product experiences a decline in demand, cash inflows are immediately affected, worsening liquidity. Similarly, incorrect procurement decisions for these high-value items can lead to disproportionately large financial losses. This high degree of dependency makes the overall business model vulnerable to demand fluctuations and inventory misjudgments, highlighting the need for tighter control over purchasing and stock management for A-class products.

3.8 Inventory Growth Over Time

Between 29 September and 3 November 2025, total inventory value increased sharply from ₹1.54 lakh to ₹53.35 lakh, representing a rise of 247% within only 35 days. Such rapid growth suggests that procurement decisions were not driven by actual sales demand but rather by forward stocking or bulk purchasing without adequate forecasting. The absence of structured reorder thresholds and demand-linked planning indicates weak inventory controls. Instead of inventory levels adjusting in response to sales performance, purchasing activity appears to be leading sales rather than following it. This reversal of the normal operating sequence results in unnecessary working capital being tied up in stock, increasing storage burden and exposing the firm to liquidity stress if sales fail to materialise at the expected pace.

3.9 Cost-Tier Segmentation and Stock Risk

Figure 4: Cost Tier Distribution (Low / Medium / High Cost)

Cost Tier Distribution: High = 75.2% of Value

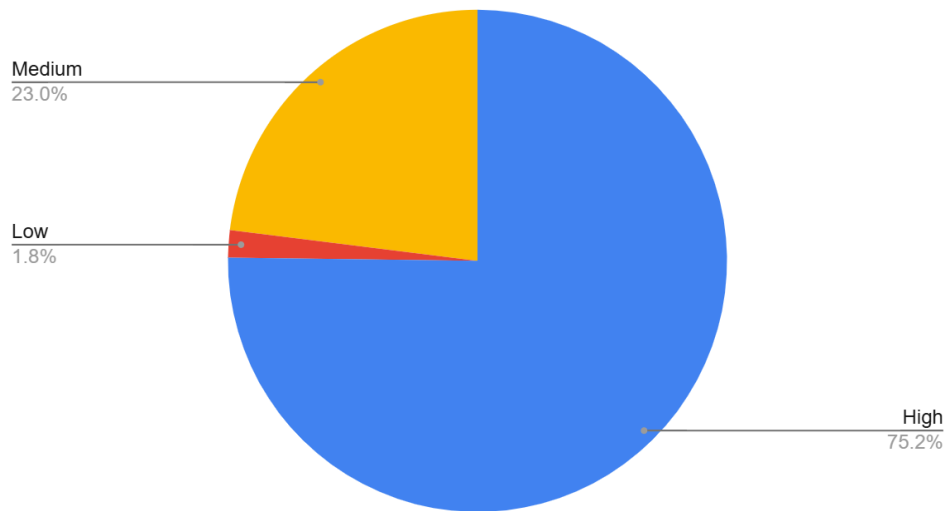
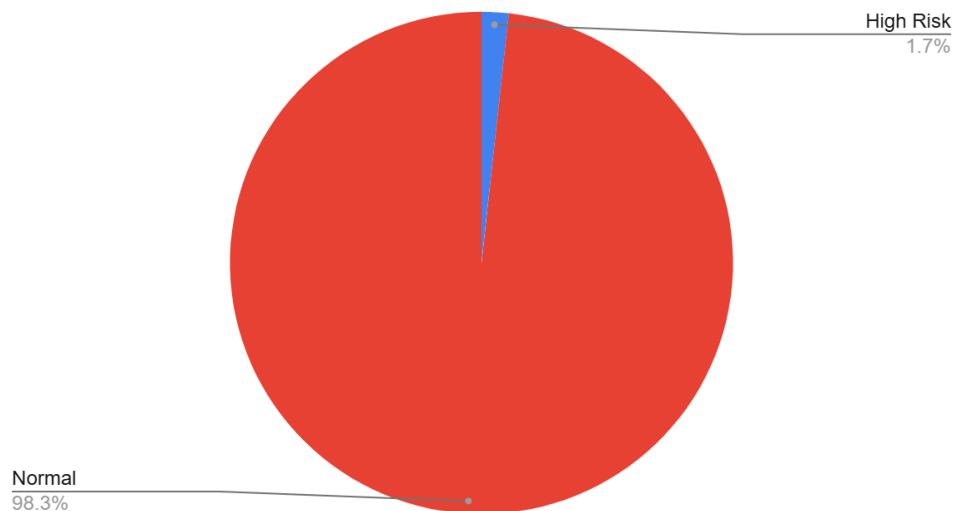


Figure 5: High-Risk Stock Classification

Stock Risk: High Risk = ₹92K (1.7%)



Cost-tier analysis shows that high-cost items dominate total inventory value despite representing a relatively small share of total stock volume. This concentration indicates that a large portion of the firm's working capital is locked into expensive equipment. Although such products may generate higher revenue per unit, they also create greater financial exposure if demand weakens, if items remain unsold for extended periods, or if products face expiry or

technological obsolescence. In the absence of structured risk buffers such as insurance, safety stock planning, or phased procurement, any disruption in the movement of these high-value items has immediate implications for liquidity and profitability.

Further risk is introduced by the presence of several SKUs classified as high-risk due to the combination of low physical availability and high monetary value. These items represent a dual operational vulnerability. On one hand, limited stock levels increase the likelihood of stock-outs, which can disrupt order fulfilment and damage customer relationships. On the other hand, their high cost makes unsold inventory financially burdensome, increasing capital blockage. Weak controls over such items also raise the probability of urgent replenishment at unfavourable prices, reducing cost efficiency. Together, these patterns reveal that although inventory appears diversified in quantity, financial risk remains highly concentrated in a narrow group of high-value and high-risk items.

3.10 Consolidated Findings

The analysis highlights several connected issues that are affecting the firm's financial health. Payments from customers are consistently delayed, with just four high-risk clients accounting for most outstanding dues. This has put continuous pressure on cash flows, which is reflected in a current ratio below one, showing that the company does not have enough short-term funds to comfortably meet its obligations. At the same time, inventory value is concentrated in only three major products, making the business highly dependent on a small number of items. Inventory also grew sharply—by 247 percent within just 35 days—indicating that purchases were made without considering realistic sales demand.

Together, these findings point to a clear imbalance between money coming in and money going out. While funds are being spent quickly on stock, payments from customers remain delayed, leading to ongoing strain on liquidity. The issue is not limited to one department but reflects an overall lack of financial control, where credit policies are not strictly followed, purchasing decisions lack planning, and monitoring happens only after problems arise. Unless corrective action is taken, this imbalance is likely to worsen and increase financial risk. These findings form the basis for the recommendations presented in the next section.

4. INTERPRETATION OF RESULTS AND RECOMMENDATIONS

This section interprets the findings of the analysis in relation to Ace and Bees Healthcare Devices' operational challenges and suggests practical measures to improve cash flow, reduce financial risk, and control inventory more effectively.

Interpretation of Results

The analysis clearly shows that delayed payments are not occasional issues but part of a larger credit-management weakness. A small number of clients are responsible for most overdue invoices, and in several cases payments have been delayed for more than six months. This indicates that customers are being allowed to extend credit without strict monitoring or consequences. As a result, the firm is unintentionally financing its customers by allowing long, interest-free payment periods while absorbing the cost of delays. Over time, this restricts the firm's ability to meet expenses, invest in stock wisely, or plan for expansion.

Inventory patterns further intensify this financial pressure. A large portion of inventory value is concentrated in just a few high-priced products, increasing vulnerability to demand fluctuations. The sharp rise in inventory value over a short period—247 percent in just over a month—suggests that purchasing decisions were not closely linked to actual sales demand. Instead of inventory growing in response to sales, stock accumulation appears to be speculative, tying up large amounts of money in unsold goods. This creates a situation where cash remains blocked in inventory while payments from customers are also delayed.

Together, these issues indicate a deep imbalance between cash inflows and capital usage. The core problem is not low sales but weak financial discipline. Funds continue to be spent on stock while payments remain unpaid, creating constant liquidity pressure. Without change, this pattern is likely to worsen and may eventually threaten operational stability.

Recommendations and Expected Impact

The most urgent priority is strengthening credit control. High-risk clients should be required to make advance payments or follow much shorter credit terms, while medium-risk clients should be regularly reviewed. New credit should be granted based on past payment behaviour

rather than order size alone. At the same time, recovery efforts should focus on invoices that are overdue by more than sixty days, starting with high-value accounts.

Inventory management also requires correction. High-value products should be monitored closely, with clear limits on how much stock can be purchased without confirmed demand. Slow-moving or low-value items should be cleared through discounts or bundled offers to free up working capital. Purchases going forward must be aligned with sales trends rather than assumptions.

To make these changes sustainable, the company could adopt a system of monthly monitoring using dashboards that track receivables, aging patterns, and inventory levels. This approach could help management identify risks earlier and take corrective action before problems become severe.

If implemented, these measures are likely to improve cash collection, could reduce exposure to unpaid invoices, and may prevent excessive investment in inventory. Over time, this may strengthen liquidity, lower overall business risk, and support more controlled growth.